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Ministry of Planning, Economic Development & International Cooperation

From the People of Japan

UNDP

Digital Agribusiness Map:

Agri Tech & Climate-Smart Agriculture in Egypt

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This Digital Agribusiness Map has been developed with funding from the Government of Japan to help make Egypt's communities and agri-food systems more resilient.

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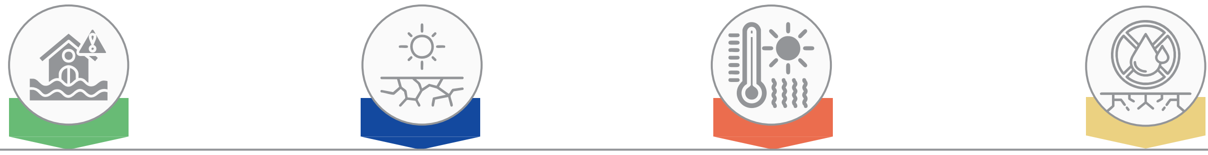
Egyptian
Agri-Business Overview



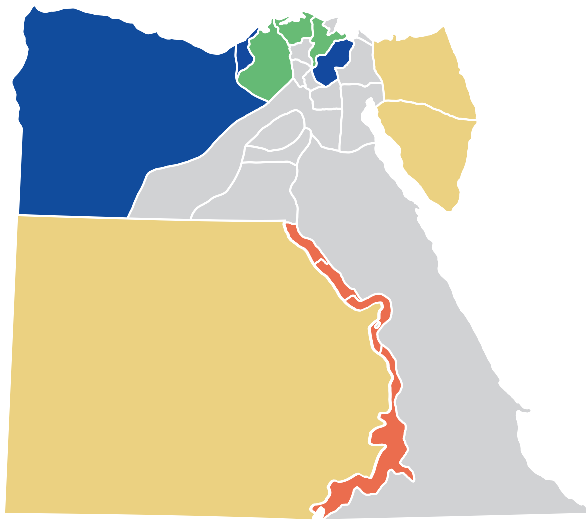
Salinity & Flooding | **Drought & Seawater Intrusion** | **Heatwave & Drought** | **Water Scarcity & Heat**

Egypt's agriculture sector is under mounting pressure, facing systemic vulnerabilities shaped by climate change, resource scarcity, market inefficiencies, and barriers to financial access. Agriculture remains a vital pillar of the Egyptian economy, employing approximately 27% of the national workforce and contributing around 11% of GDP (CAPMAS, 2024). However, the sector's productivity is constrained by a narrow arable land base—less than 4% of Egypt's total land area—combined with declining water resources, outdated practices, and uneven access to markets and finance.

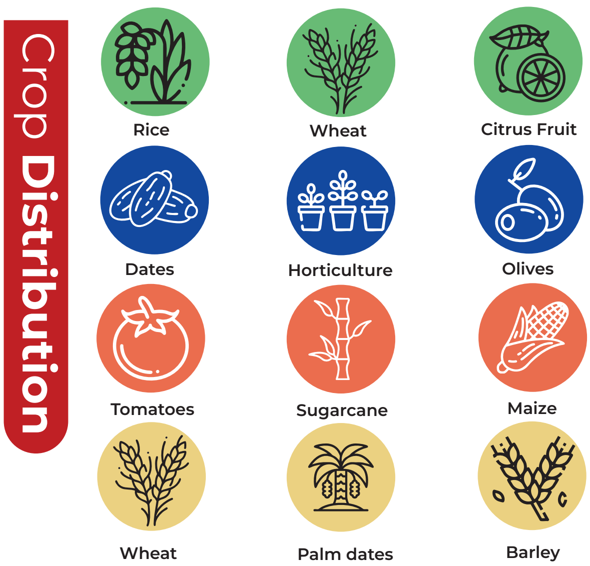
Climate change alone is projected to reduce Egypt's agricultural productivity by up to 18% by 2050 (IFPRI, 2023), with wheat and maize yields already showing significant sensitivity to rising temperatures and erratic rainfall patterns. Meanwhile, per capita freshwater availability has dropped from 900 m³/year in 2000 to 480 m³/year by 2023, placing Egypt well within the water poverty threshold (Ministry of Water Resources, 2023).

Agricultural modernization, digital transformation, and climate adaptation are thus not optional but necessary for Egypt's food security and economic resilience. However, these dynamics vary significantly across different regions of the country, depending on ecological conditions, market access, technological infrastructure, and socio-economic factors. A one-size-fits-all approach will not succeed. A regionally tailored, evidence-based strategy is essential.

This report, developed as part of the UNDP-supported program, provides a granular, regional analysis of Egypt's agribusiness landscape, intended to serve as a foundational input for the Digital Agribusiness Map initiative.

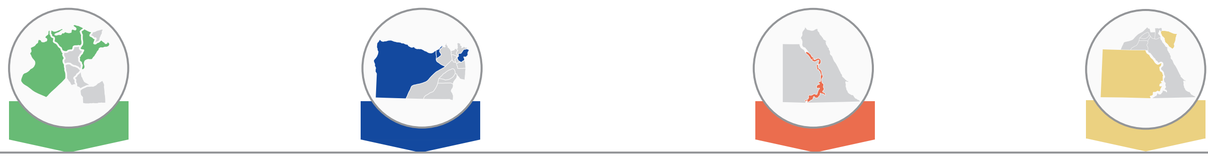


Delta **Region** | Northern Coastal **Region** | Upper Egypt **Region** | Desert **Region**



Explore the Egyptian Agri-Business Regions

Explore the **Methodology**



Delta Region | **Northern Coastal Region** | **Upper Egypt Region** | **Desert Region**

Delta Region
(Kafr El Sheikh, Dakahlia, Beheira)

Climate Challenges	Startups Solution Challenges	Crop Distribution	Agri-tech Adoption	Farmer Financial inclusion
			15%	20%
Salinity & Flooding	**Smart Irrigation**	**Rice, Wheat, & Citrus Fruit**	**Precision Irrigation Adoption**	**Access to Loans, Limited Micro finance Penetration**

The Delta Region — encompassing Kafr El Sheikh, Dakahlia, and Beheira — constitutes the agricultural heartland of Egypt, contributing approximately **35% of national food production** (CAPMAS, 2024). With its historically fertile alluvial soils and extensive irrigation networks fed by the Nile, the Delta has traditionally served as Egypt's breadbasket, producing key staples such as rice, wheat, citrus fruits, and vegetables.

However, the region is now at the forefront of Egypt's agricultural vulnerability to climate change, soil degradation, and water resource stresses. Increasing salinity, groundwater mismanagement, pest proliferation, and extreme weather events are putting the Delta's productivity and sustainability at serious risk. Without significant adaptation interventions, Egypt risks losing a major share of its domestic food supply, exacerbating import dependence and rural poverty.

Recent projections from the Ministry of Agriculture (2024) estimate that unless comprehensive resilience strategies are implemented, the Delta could experience a **15–20% decline** in agricultural output by 2035. Moreover, given the region's high population density — with over **40 million** inhabitants residing within the broader Delta catchment — disruptions to agriculture have direct socio-economic implications, including rural-urban migration, food price inflation, and employment instability.

Recognizing the Delta's strategic importance, this section maps the key climate challenges, entrepreneurial innovations, crop dynamics, technology adoption patterns, and market access structures shaping the region's agricultural future.

Delta **Region**

Startups **with Solutions**

Baramoda
Soil Health & Salinity Management

Mozare3
Digital Farmer Financing & Contracting

NileBot
Smart Irrigation Management

Agrimatic Farms
Hydroponic & Controlled- Environment Agriculture

FreshSource
Agri-Supply Chain Optimization

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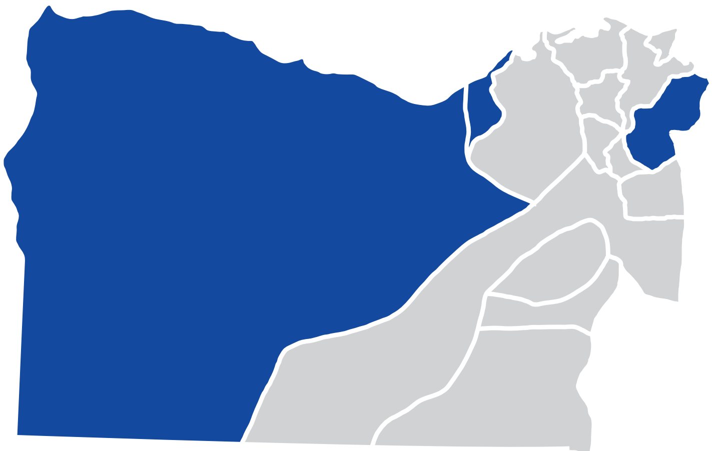
Northern Coastal Region
(Alexandria, Matrouh, Port Said)

Climate Challenges	Startups Solution Challenges	Crop Distribution	Agri-tech Adoption	Farmer Financial inclusion
			10%	25%
Drought & Seawater Intrusion	**Water Saving Technology**	**Dates, Horticulture, & Olives**	**Adoption of Hydroponic Tech**	**Mobile Payments; Limited Insurance**

The Northern Coastal Region of Egypt —comprising Alexandria, Matrouh, and Port Said—forms a distinct agricultural and ecological zone marked by Mediterranean climatic influences, strategic port infrastructure, and a rising focus on horticultural export crops such as olives, dates, figs, and medicinal herbs. Although it contributes a smaller share of total national agricultural output **(around 9–12%, CAPMAS 2024)**, the region is a strategic node in Egypt's agribusiness value chain due to its export orientation and growing AgriTech experimentation zones in peri-urban Alexandria.

Coastal governorates are increasingly vulnerable to compounded climate risks: drought cycles, shifting rainfall patterns, rising sea levels, saltwater intrusion, and aridification of inland plains. In Matrouh, for instance, erratic precipitation has led to chronic water scarcity and land abandonment, while in Port Said, water tables are at risk of salinization from sea encroachment. These trends severely constrain smallholder farming livelihoods, particularly among Bedouin communities and informal landholders who rely on rain-fed agriculture or shallow wells.

Furthermore, infrastructure bottlenecks, including cold chain limitations, high last-mile transport costs, and port logistics delays, undermine market access for high-value crops. Despite this, Alexandria is emerging as a secondary innovation hub, with startups experimenting with hydroponics, water-efficient systems, and post-harvest logistics technologies. The Coastal Region presents both a climate fragility frontier and an innovation opportunity zone, especially for Egypt's horticulture and export diversification strategies.



Northern Coastal **Region**

Startups **with Solutions**

Dessaal
Water Efficiency & Drip Irrigation

Baramoda
Soil Health Restoration

Hydrofarms
Hydroponics & Controlled Environment Agriculture

Yalla Mozare3
Digital Extension & E-Commerce

GoAgri
Climate Risk Advisory

Overview | **Delta Region** | **Upper Egypt Region** | **Desert Region**

Policy Recommendations | **Learn More**

Upper Egypt Region
(Minya, Sohag, Qena, Aswan)

Climate Challenges	Startups Solution Challenges	Crop Distribution	5%	15%
Heatwave & Drought	**Climate-Resilient Seeds**	**Tomatoes, Sugarcane, & Maize**	Agri-tech Adoption	Farmer Financial inclusion **Micro finance**

Upper Egypt spans a predominantly arid zone along the southern Nile Valley, encompassing governorates like Minya, Sohag, Qena, and Aswan. Though often marginalized in national policy and investment flows, Upper Egypt is a vital agricultural region — accounting for approximately **27% of Egypt's** cultivated land and housing **over 20 million people**, most of whom depend directly on agriculture for income (CAPMAS, 2024).

The region specializes in sugarcane, tomatoes, maize, and horticultural crops, particularly in Minya and Qena. It also produces a growing share of Egypt's winter vegetables and supports important livestock and dairy economies. However, it remains deeply vulnerable to climate stressors, water scarcity, and infrastructure deficits — exacerbated by high poverty rates, weak financial services, and limited AgriTech access.

Upper Egypt faces the highest rates of rural poverty in the country, with **more than 45%** of agricultural households living under the poverty line (World Bank, 2023). Infrastructural isolation and governance fragmentation have compounded climate impacts, particularly in areas such as Aswan, where desertification and extreme temperatures are intensifying.

Despite these challenges, Upper Egypt holds huge potential for climate-smart agriculture, especially through the expansion of resilient seed systems, drip irrigation, solar-powered farming, and agro-processing clusters. This section maps the five primary climate-related challenges affecting agriculture in Upper Egypt.

Upper Egypt **Region**

Startups **with Solutions**

Shoura Seeds
Climate-Resilient Seed Development

MwaslaTech
Digital Extension Services

Tadweer Agriculture
Organic Waste & Soil
RehabilitationEnvironment

IrriSat
Satellite- Supported Irrigation Planning

Desert Region
(New Valley, Toshka, Sinai)

Climate Challenges	Startups Solution Challenges	Crop Distribution	Agri-tech Adoption	Farmer Financial inclusion
Water scarcity & Extreme Heat	**Drip Irrigation & Desert Farming**	**Wheat, Palm Dates, & Barley**	3% **Precision Irrigation Pilots**	10% **Government Subsidies**

The Desert Region — comprising the New Valley, Toshka, and parts of Sinai — represents Egypt's frontier for agricultural expansion under conditions of extreme aridity and climate vulnerability. These areas are central to the government's land reclamation strategy, particularly under national mega projects such as the **1.5 Million Feddan** Project and the Toshka Revival Initiative. While these projects seek to increase agricultural output and reduce dependence on the Nile Valley, the harsh ecological conditions, infrastructural gaps, and weak economic integration of desert farming zones pose significant challenges.

Unlike the fertile Nile Delta or the floodplain corridors of Upper Egypt, desert agriculture is dependent on deep groundwater aquifers, solar-powered irrigation, and artificial soil amendments. These regions are characterized by extremely low rainfall **(<50 mm/year)**, high evapotranspiration rates, and wide temperature fluctuations between day and night — factors that place immense strain on both crops and infrastructure. In Sinai, security constraints and land tenure complexities further inhibit investment and scaling.

According to the Ministry of Agriculture (2024), desert governorates together account for only 8–10% of national agricultural production, despite comprising over **90% of Egypt's** total land mass. This stark contrast highlights both the untapped potential and the structural constraints facing desert agriculture. Yet, the region is also a critical testbed for innovation, particularly in precision irrigation, desert-adapted crop varieties, and solar-powered water systems. If scaled effectively, successful models from the desert frontier could inform sustainable agricultural transitions across arid regions globally.

Desert **Region**

Startups **with Solutions**

DesertAgro
Precision Drip Irrigation & Remote Sensing

Baramoda
Soil Regeneration for Arid Zones

WASDEL
Low-Energy Water Purification for Irrigation

Palmatec
Date Palm Optimization & Disease Control

ReefTech
Off-Grid Agri- Energy Systems

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